United International University

Department of Computer Science & Engineering

CSE 3715: Data Communication

Class Test 03

Duration: 30 Minutes | Total Marks: 20

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1. A sender wants to transmit the dataword:

$$D(x) = x^6 + x^4 + x^2 + 1$$

using the generator polynomial:

$$G(x) = x^3 + x + 1$$

(a) Determine the **encoded message** that will be transmitted using the CRC technique. **[7]**

(b) During transmission, the **3rd bit from the left** of the transmitted message is flipped. Calculate how the receiver detects the error. **[7]**

2. The receiver gets the following five 8-bit blocks (including checksum):

$$10101100 \quad 11001010 \quad 11110000 \quad 01100111 \quad 01001101$$

Determine whether any **error occurred** during transmission. **[6]**

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